

AI Tool Usage & Costs

NZIPL Clean Value Chain Explorer — March–June 2026

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Purpose

This document summarises AI and software tool subscriptions used during the development of the **NZIPL Clean Value Chain Explorer (CVCE)** and related research, covering **March 6 – June 1, 2026**. It is intended to support cost-sharing discussions between personal and institutional accounts, and to document the transition to a stable workflow configuration.

Current Workflow Setup (May 2026)

The subscription stack has stabilised into two distinct tiers:

Tier	Tool	Purpose
NZIPL	Claude Code (Team Pro)	Primary AI coding environment for all CVCE/NZIPL work — reads/writes R, JS, Python; manages git; builds data pipelines; debugs and edits interactive dashboards
Personal	DeepSeek API	Cost-efficient API for personal research and scripting tasks outside NZIPL scope
Personal	Qwen (local model)	Offline personal assistant; zero marginal cost
Personal (occasional)	Claude Pro	Occasional personal queries via web interface

ChatGPT Plus — cancelled April 2026. No longer in the subscription stack. Usage was consolidated into Claude, which handles all the tasks ChatGPT was previously used for (academic writing, reasoning cross-checks, equation formatting).

Cost Summary: Lab Period (August 15, 2025 – June 1, 2026)

Service	Period	Months	Per month	Total (EUR/USD)
Claude / Claude Code (Anthropic)	Mar 6 – Jun 1, 2026	~3	EUR 18–46/mo	EUR 427.60 ¹
ChatGPT Plus (OpenAI)	Aug 17 – Apr 17, 2026	9 ¹	\$21.63	\$194.67
Windsurf Pro (Exafunction)	Sep 4 – Mar 4, 2026	7	\$16.22	\$113.54
Posit Connect Cloud	Sep 27 – Feb 27, 2026	6	\$20.54	\$123.24
DeepSeek API	May 2026 –	ongoing	pay-as-you-go	Personal only
Qwen (local)	May 2026 –	ongoing	\$0	Personal only
TOTAL (in- stitutional scope)				\$885²

¹ EUR 427.60 confirmed final (14 invoices, Mar 6 – May 11, 2026; verified via Gmail). May 28–Jun 1 sessions ran on existing prepaid credit — no additional invoices issued. Lab period ChatGPT only (Aug 2025 – Apr 2026); pre-lab months (Jan–Jul 2025) excluded as personal. ² EUR 427.60 @ 1.08 USD/EUR = \$461.81 + \$431.45 USD tools = ~\$893. No further charges after May 11.

Anthropic invoice detail (all receipts)

Invoice	Date	Amount	Type	Project
NJJAFZLE-0001	Mar 6	EUR 21.78	Claude Pro – month 1	CVCE setup
NJJAFZLE-0002	Mar 23	EUR 6.05	Extra usage	LEEDS_MODEL
NJJAFZLE-0003	Mar 31	EUR 6.05	Extra usage	LEEDS_MODEL
NJJAFZLE-0004	Apr 1	EUR 24.20	Extra usage	LEEDS_MODEL
NJJAFZLE-0005	Apr 2	EUR 24.20	Extra usage	LEEDS_MODEL (submission)
NJJAFZLE-0006	Apr 2	EUR 24.20	Extra usage	LEEDS_MODEL (submission)
NJJAFZLE-0007	Apr 6	EUR 21.78	Claude Pro – month 2	CVCE
NJJAFZLE-0008	Apr 8	EUR 46.28	Team Pro prepaid	CVCE
NJJAFZLE-0009	Apr 15	EUR 46.28	Team Pro prepaid	CVCE

Invoice	Date	Amount	Type	Project
NJJAFZLE-0010	Apr 28	EUR 46.28	Team Pro prepaid	CVCE + CONSERVATION
NJJAFZLE-0011	Apr 29	EUR 46.28	Team Pro prepaid	CVCE + CONSERVATION
NJJAFZLE-0012	May 6	EUR 21.78	Claude Pro – month 3 (personal)	Personal
NJJAFZLE-0013	May 7	EUR 46.28	Team Pro prepaid	CVCE Phase II
NJJAFZLE-0014	May 11	EUR 46.28	Team Pro prepaid	CVCE Phase II
Total		EUR 427.60		

May 2026 additions: EUR 114.34 (invoices 0012–0014) — reflecting continued intensive Phase II development (Generic Country Atlas builder, Global Atlas, panel system, color mode toggle, SHAP-RCA scatter).

Project Attribution

Based on git commit history and session logs through May 20, 2026:

Project	Claude share	Claude (EUR)	Evidence
CVCE / NZIPL	~72%	~308	Phase I (Apr): Global Atlas, Chile/Brazil Atlases, Intelligence Report. Phase II (May): Generic Atlas builder, 7-country atlases, 13 Global Atlas sheets, category drums, color mode toggle, SHAP-RCA scatter, data summaries
LEEDS_MODEL	~15%	~64	8 commits Apr 2 (submission sprint); zero after Apr 3

Project	Claude share	Claude (EUR)	Evidence
CONSERVATION papers	~8%	~34	Files modified Apr 22–30 (sraffa_acs, econ_thermo, conserva- tion_PhysicaA)
Personal (misc.)	~5%	~22	May 2026 personal queries via Claude Pro (NJJAFZLE-0012)
Total	100%	427.60	

Activity timeline

Period	Activity	Project
Mar 6–22	Claude Pro starts; workplan, render scripts, deploy setup	CVCE
Mar 23 – Apr 3	MRIO equations, paper editing, submission sprint	LEEDS_MODEL
Apr 3	<i>Paper submitted to Ecological Economics SI</i>	—
Apr 8–20	type→role rename; Global Atlas v1; Chile/Brazil Atlases	CVCE
Apr 20–27	Intelligence Report; ecoclassical.github.io restructure	CVCE
Apr 22–30	Conservation papers parallel with CVCE	CVCE + CONSERVATION
Apr 28–30	ChatGPT cancelled; Windsurf ended; stack simplified	—
May 1–11	Generic Country Atlas (all countries × 10 techs); Global Atlas fixed; Phase II panels	CVCE
May 11–20	SHAP-RCA scatter (bug fixes); category drum + color toggle; legend redesign	CVCE
May 28–29	v2.1.0: PC scatter redesign [8,40] bubbles; 420 standalone panels (6 countries); category highlighting all atlas panels; global atlas JS scope + deploy path bugs fixed	CVCE

Period	Activity	Project
Jun 1	Global atlas: nodeColor global scope fix (Sankey/Dendrite crash); 2-col legend; full role names; selCat→map (getCatPV); activity map category filter; all 7 atlases redeployed	CVCE

Subscription Status (May 20, 2026)

Service	Status
Claude Code Team Pro (Anthropic)	Active — primary NZIPL tool; prepaid usage billing
Claude Pro (Anthropic, personal)	Active — occasional personal use only
DeepSeek API	Active — personal research; pay-as-you-go, negligible cost
Qwen (local model)	Active — personal; zero cost
ChatGPT Plus (OpenAI)	Cancelled April 2026 — replaced by Claude
Windsurf Pro	Cancelled March 2026
Posit Connect Cloud	Cancelled February 2026

Productivity Impact

Claude Code (adopted March 6, 2026) replaced a four-tool stack (ChatGPT + Windsurf + Posit + manual workflows). Git commit frequency increased **5.8×** **within the first eight weeks** and has remained elevated through May 2026.

Repository	Before Claude Code	After Claude Code	Increase
CVCE / NZIPL	6.4 / week	21.5 / week	+3.4×
LEEDS_MODEL	0.6 / week	3.6 / week	+6.0×
All repos combined	7.5 / week	44.0 / week	+5.8×

Before: Aug 15, 2025 – Mar 5, 2026 (28 weeks). After: Mar 6 – Jun 1, 2026 (~13 weeks).

Notable Phase II deliverables (May–June 2026, NZIPL scope):

- **Generic Country Atlas** — single `build_country_atlas.R` covering all countries × 10 techs
- **7 country atlases** deployed (Brazil, India, Japan, South Korea, Vietnam, Mexico, Spain)
- **13 Global Atlas sheets** with SHAP-RCA scatter panel
- **420 standalone panel files** (6 countries × 10 techs × 7 module types) rebuilt with redesigned PC scatter

- **Category-aware drum wheels** — Stage → Role → Category → Product cascade filter; wired to all panels including trade map and activity map
- **Color mode toggle** — global SR / Category coloring across all panels including timeline (via `srToCatMap`)
- **2-column SR legend + HS Category legend** — full role/category names, compact 2+1 header layout
- **Global atlas parity** — SHAP-RCA scatter, Trade Timeline, color mode toggle, Category drum matching country atlas layout

i Note

PDF receipts on disk (invoices 0001–0009, Mar 6 – Apr 15): archived in `docs/receipts/pdf/`.

Missing PDFs — download from Gmail (`invoice+statements@mail.anthropic.com`). All 14 invoices confirmed; amounts already in ledger. Only the PDF files themselves are missing for 0010–0014:

Invoice	Date	Amount	Filename to save
0010	Apr 28	EUR 46.28	<code>Anthropic_2026-04-28.pdf</code>
0011	Apr 29	EUR 46.28	<code>Anthropic_2026-04-29.pdf</code>
0012	May 6	EUR 21.78	<code>Anthropic_2026-05-06.pdf</code>
0013	May 7	EUR 46.28	<code>Anthropic_2026-05-07.pdf</code>
0014	May 11	EUR 46.28	<code>Anthropic_2026-05-11.pdf</code>

No further charges after May 11 — May 28–Jun 1 sessions consumed existing prepaid credit. Full ledger: `docs/receipts/subscription_costs.csv`.